

BIG SYNTAX DOESN'T WANT YOU TO KNOW

MAL is universal? Hardly.

The Menzerath–Altmann law on all 180 UD treebanks

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HOW WE COMPUTE MAL

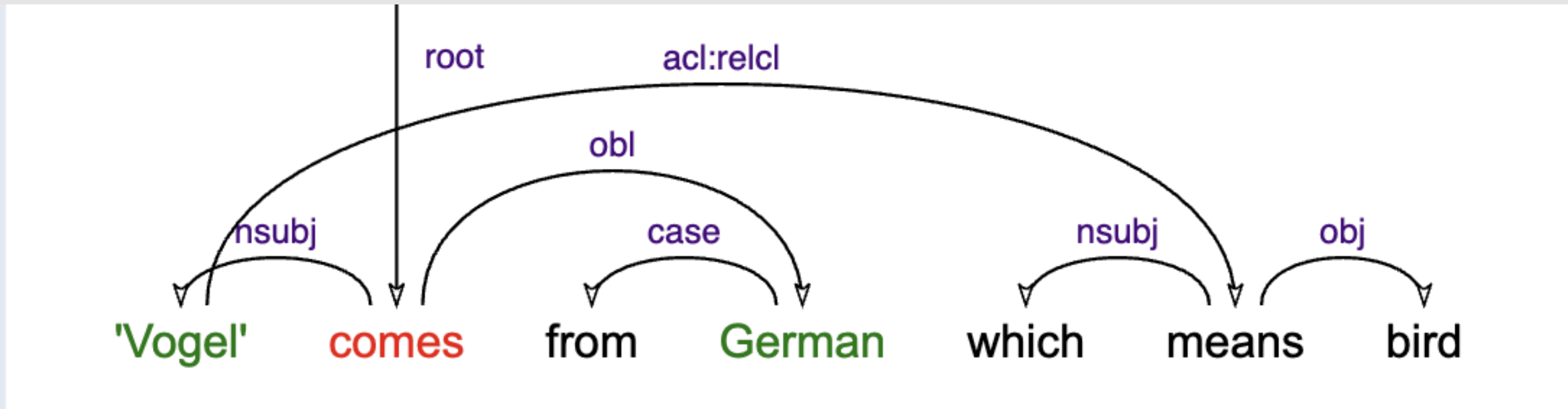
Real constituents only. Fake dependents eliminated!

The **Menzerath–Altmann law** (MAL) predicts that “the greater the number of components of a given unit, the shorter their respective length” (Menzerath, 1928). In the verbal domain: as the number of dependents n grows, their average size decreases.

Altmann (1980) formalised this as a **power law**: $MAL_n \approx n^{-\beta}$, i.e. $\log(MAL_n) \approx -\beta \cdot \log(n)$. The slope β measures the **MAL effect**. Previous studies: Macutek et al. (2017) for Czech; Tanaka-Ishii (2021) for UD 2.3 without filtering.

NEW! Our data extraction is filtered: we exclude punct, discourse, parataxis, conj, cc, vocative, aux, compound:prt, mark, case from the verbal construction. Only **real syntactic constituents** are retained.

What about “bastards”? A constituent’s subtree may spill across the verb. We count the spilled part as a separate dependent of the verb:



The relative clause “which means bird” (**acl:relcl**) syntactically depends on ‘Vogel’ (left of verb) but appears right of verb – a **bastard**. It is counted as a right-side constituent of **comes**. Source: en_gum.

NEW! LEFT VS. RIGHT

Two sides of the verb, two different worlds

We compute three variants of the MAL effect:

MAL

$MAL_n(L)$ = mean size of constituents in all verbal constructions of L with n constituents (left + right combined). Standard MAL as in the literature.

LMAL & RMAL **NEW!**

$LMAL_n(L)$ = mean size of constituents in constructions with n **preverbal** (left) constituents, regardless of the right side.

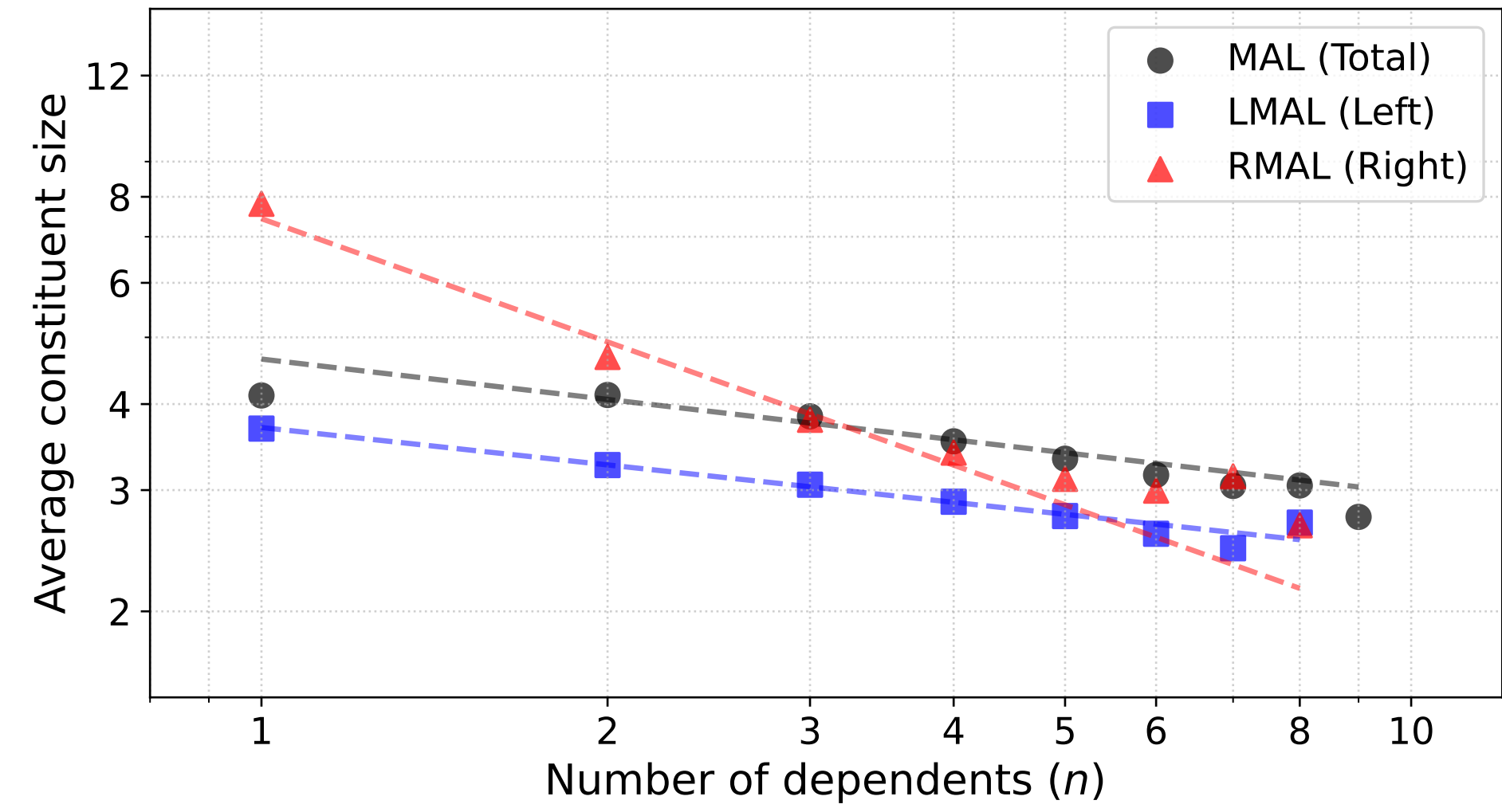
$RMAL_n(L)$ = mean size of constituents in constructions with n **postverbal** (right) constituents, regardless of the left side.

MAL effect = β , the slope of the regression line in log-log space. $\beta > 0.1$: **MAL language**. $\beta < -0.1$: **Anti-MAL**. In between: **grey zone**.

Why does this matter? DLM (Dependency Length Minimization) predicts short-before-long postverbally and long-before-short preverbally. Unlike DLM, MAL is direction-free. So: **does MAL hold equally on both sides?** Spoiler: **no**.

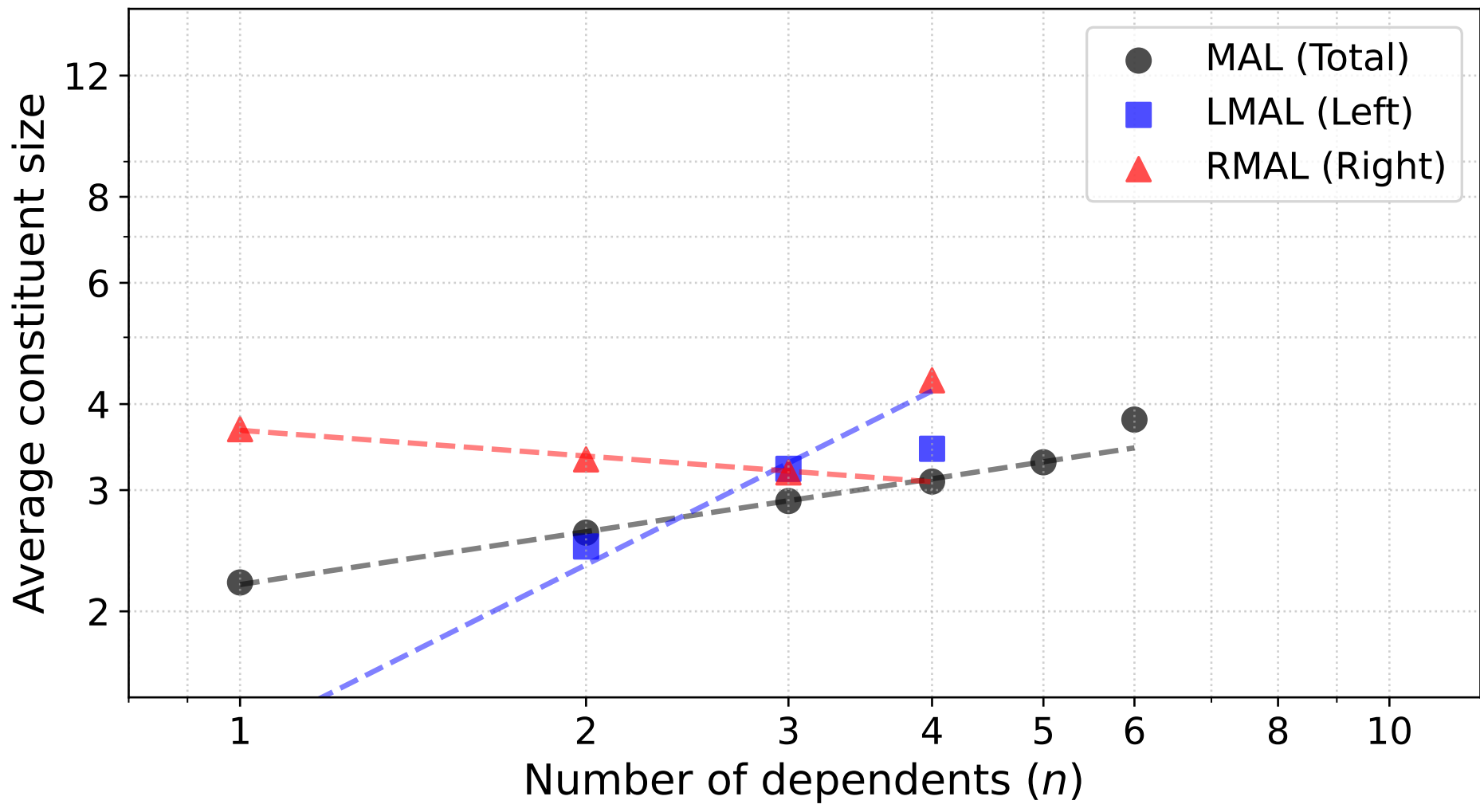
AS USUAL

German nails it. Naija wobbles. Arabic surprises. You?



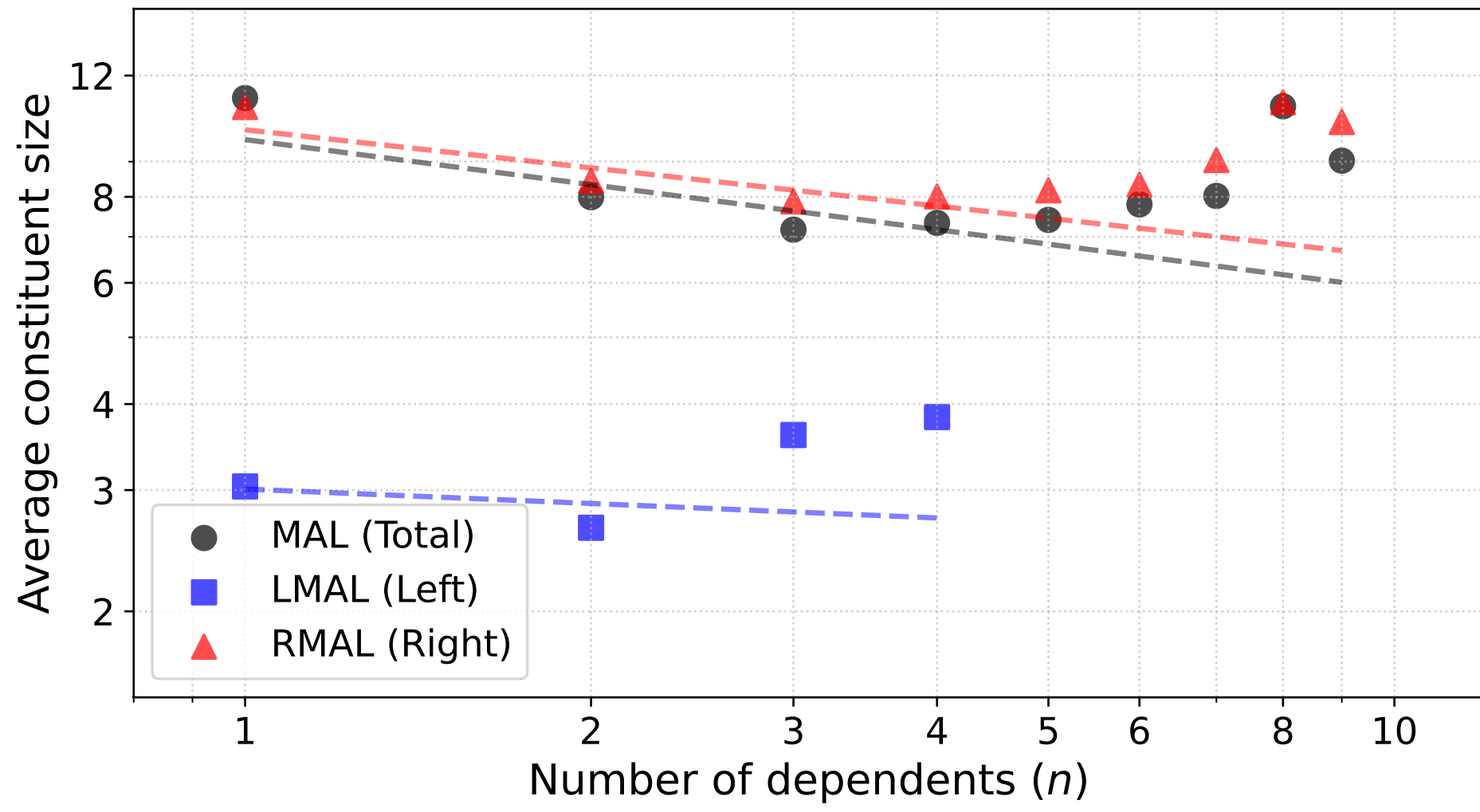
German – textbook MAL

slope = -0.17, R^2 ~ 0.89 on the log-log fit. A near-perfect power law: every extra dependent shrinks the average constituent. German is MAL, RMAL and LMAL as expected. Is anyone surprised?



Naija – mixed, asymmetric

Left and right dependents diverge wildly. A young pidgincreole refusing to fit one curve – in fact the **strongest Anti-LMAL in the whole sample** ($\beta = +0.73$ preverbally).

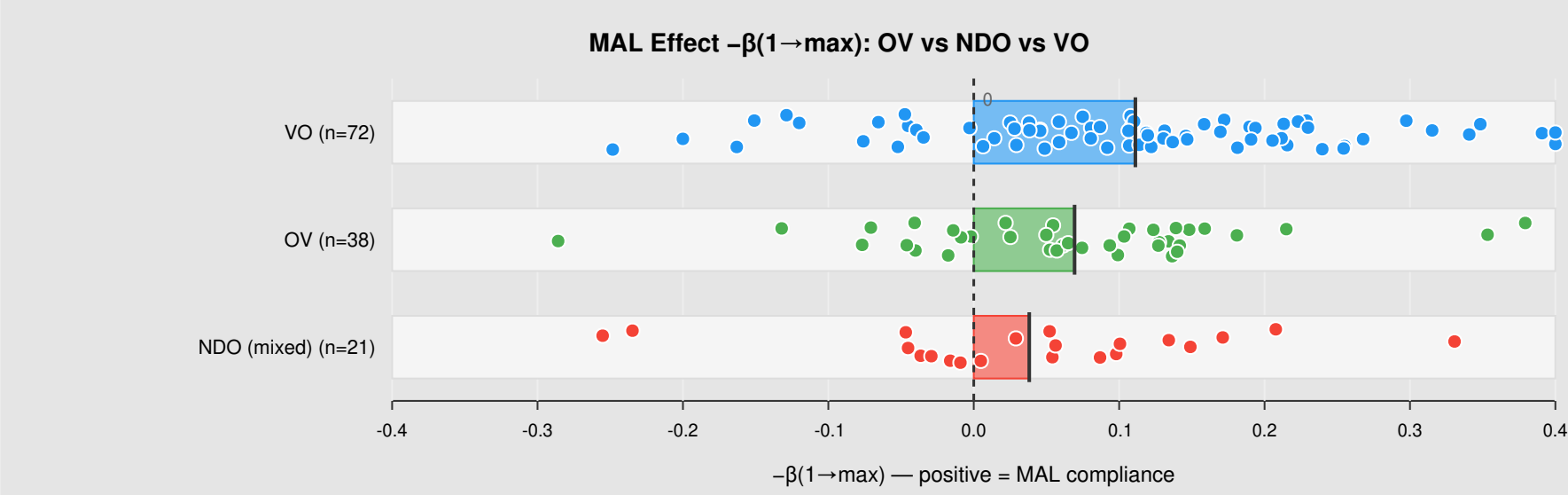


Arabic – complex behavior

Arabic has a particularly low R^2 . It follows MAL from $n = 1$ to 3, but becomes Anti-MAL from $n = 3$ onwards. **And you?** Check your language at typometrics.elizia.net!

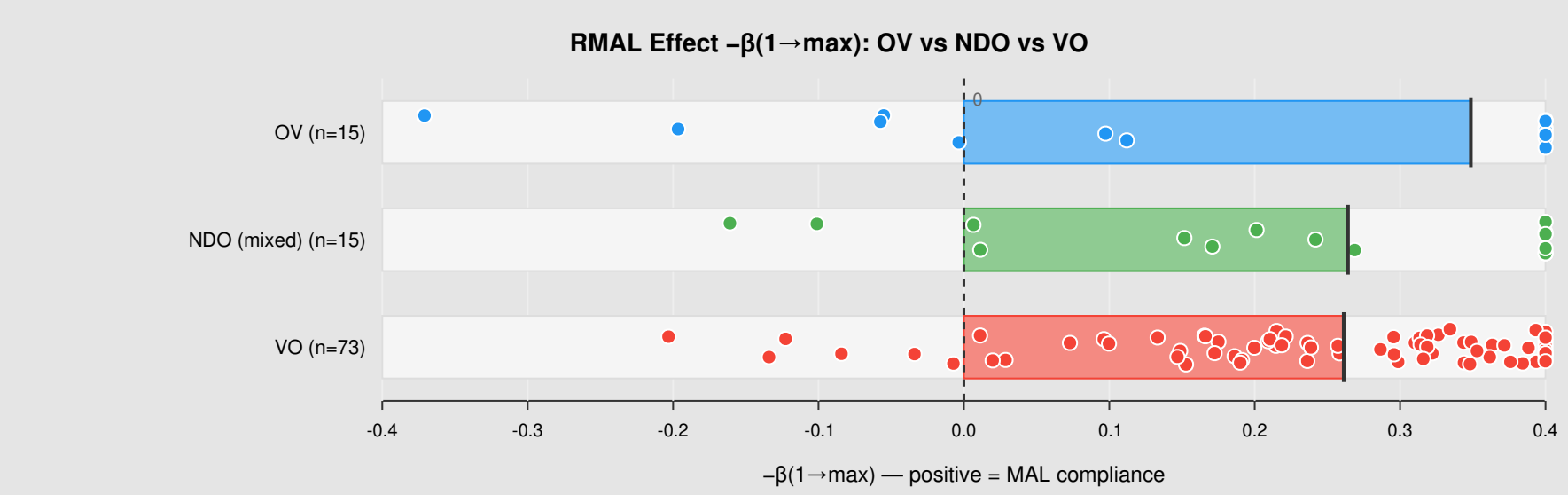
THE NUMBERS NOBODY TALKS ABOUT

MAL is not ambidextrous. The verb leans right.



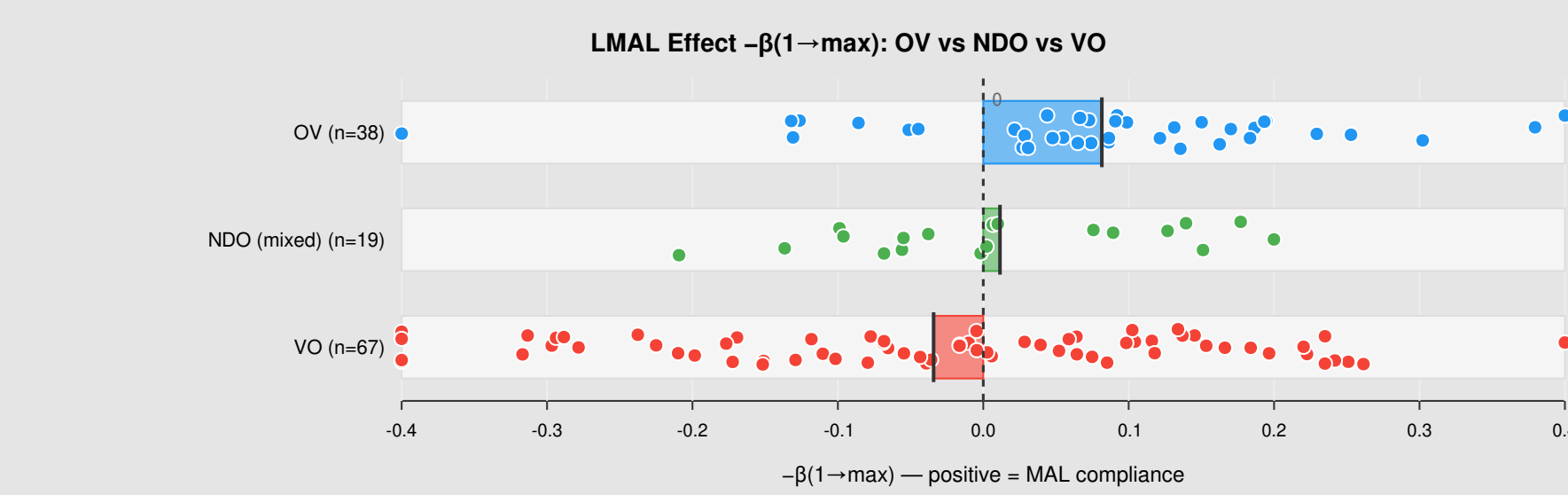
MAL by word order

MAL is stronger for **VO languages** (56%) than OV (42%), but RMAL and LMAL are **stronger for OV!** Where does the paradox come from?



RMAL: postverbal domain

79% of UD languages are RMAL (81/103) and only **7% are Anti-RMAL** (7/103). The right side of the verb strongly compresses constituent size as n grows. VO languages lead: 84% are RMAL vs. 63% for OV.



LMAL: preverbal domain

Only **31% are LMAL** (39/124) and **23% are Anti-LMAL** (29/124). The left side of the verb resists compression – or even reverses it. OV languages are better: 39% are LMAL vs. 29% for VO. VO languages are **3×** more likely to be Anti-LMAL.

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